

AUTO LOAN SECURITISATION ANALYTICS USING POWER BI

Data Analyst Internship Project

Tools Used

- Power BI
- Power Query
- DAX
- CSV Datasets

Project Domain

Structured Finance & Credit Risk Analytics

Project Overview

Overview

Auto Loan Securitisation Analytics is a business intelligence solution developed using Power BI to monitor portfolio performance, credit risk, delinquency trends, expected losses, investor reporting, and stress testing.

The dashboard provides a centralized platform for analysing securitised auto loan portfolios and supports risk management decision-making through interactive visualizations and KPI tracking.

Business Objectives

Project Objectives

- Monitor portfolio exposure and balances
- Track delinquency and default behaviour
- Measure expected credit losses
- Support IFRS 9 reporting requirements
- Analyse vintage performance
- Perform stress testing
- Simulate securitisation waterfalls
- Support investor reporting

Expected Outcome

A comprehensive analytics platform for portfolio monitoring and credit risk assessment.

Dataset Description

Datasets Used

Auto Loan Securitisation Data

Contains borrower information, loan balances, interest rates, credit scores, and portfolio exposure metrics.

DPD Snapshot History

Tracks monthly delinquency status and overdue amounts.

Static Pool Vintage Data

Supports vintage and cohort performance analysis.

Dynamic Loss Monthly Data

Contains monthly loss, recovery, and collection metrics.

Data Model & Architecture

Power BI Data Model

The solution follows a star-schema architecture.

Fact Tables

- Auto Loan Securitisation Data
- DPD Snapshot History
- Static Pool Vintage Data
- Dynamic Loss Monthly

Dimension Tables

- Date Table
- Waterfall Steps

Benefits

- Faster reporting
- Better scalability
- Efficient DAX calculations
- Improved dashboard performance

DAX Measures & KPIs

Key DAX Measures

Portfolio Metrics

- Total Exposure
- Net Cashflow
- Total Loans

Risk Metrics

- Average PD
- Average LGD
- Total EAD
- Total ECL

Stress Metrics

- Base Expected Loss
- Mild Stress Expected Loss
- Severe Stress Expected Loss

These measures form the foundation of portfolio monitoring and risk analysis.

Dashboards Developed

Dashboard Modules

Executive Dashboard

Portfolio KPIs and exposure analysis.

DPD Analysis Dashboard

Delinquency bucket monitoring.

Vintage Analysis Dashboard

Cohort performance tracking.

IFRS 9 Dashboard

Expected credit loss monitoring.

Waterfall Dashboard

Cash flow allocation analysis.

Stress Testing Dashboard

Scenario-based risk evaluation.

Key Results & Insights

Project Findings

Portfolio Exposure

317.79 Million

Net Cashflow

315.50 Million

Expected Loss

2.29 Million

Major Observations

- Majority of loans remain Current
- Stage 1 dominates the portfolio
- Vintage performance varies across cohorts
- Stress scenarios significantly increase losses
- Waterfall structure protects senior investors

Slide 9: Business Value

Benefits of the Solution

Risk Monitoring

Provides continuous visibility into portfolio quality.

Investor Reporting

Supports securitisation performance reporting.

Credit Risk Assessment

Measures expected losses and default risk.

Regulatory Reporting

Supports IFRS 9 impairment analysis.

Decision Support

Enables data-driven portfolio management.

Conclusion

The Auto Loan Securitisation Analytics Dashboard successfully implements a complete securitisation monitoring framework using Power BI.

Achievements

- Portfolio Analytics
- DPD Monitoring
- Vintage Analysis
- IFRS 9 Reporting
- Waterfall Modelling
- Stress Testing
- Investor Reporting

The solution provides a unified platform for monitoring portfolio performance, managing credit risk, and supporting investor decision-making.